

Assessing Spatial Accessibility of Emergency Mental Health Services in the United States

Meixian Li Contact: meixian2@Illinois.edu Mentor: Dr. Mengxi Zhang, Dr. Junghwan Kim

Department of Geography and Geographic Information Science, University of Illinois Urbana-Champaign

INTRODUCTION

122 million Americans live in designated mental health care shortage areas. To address this critical shortage of mental health providers, the United States has expanded emergency mental health services. However geographic accessibility and factors that impact accessibility of these services have not yet been well addressed.

Why Emergency Mental Health

Emergency mental health services play a critical role during acute psychiatric crises, when rapid access can prevent harm, reduce unnecessary hospitalizations, and limit prolonged emergency department use. In response to rising demand, federal and state governments have increasingly invested in community-based crisis services as alternatives to inpatient psychiatric care and hospital emergency departments. Psychiatric Emergency Walk-In Clinics and Crisis Intervention Teams (CITs) represent two complementary models of emergency mental health care. Walk-in clinics provide immediate, in-person crisis assessment, while CITs respond directly to psychiatric emergencies in the community.



Crisis Response Framework created by Substance Abuse and Mental Health Services Administration

Research Gap

Despite growing policy emphasis on these services, whether they are geographically accessible to the populations who need them most remains poorly understood. Prior research often relies on service utilization as a proxy for access, which may misrepresent true accessibility, particularly for time-sensitive emergency care.

Objective

This study uses geospatial analysis to assess nationwide accessibility to Psychiatric Emergency Walk-In Clinics and Crisis Intervention Teams across the United States. Specifically, we aim to:

1. Measure county-level accessibility to emergency mental health services within 15- and 30-minute driving-time thresholds
2. Examine geographic variation in accessibility across counties and states
3. Identify socioeconomic factors associated with accessibility, including population density, income, educational attainment, and racial composition

DATA AND METHOD

Study Design

We assessed geographic accessibility to Psychiatric Emergency Walk-In Clinics and Crisis Intervention Teams (CIT) across the United States. Accessibility was first measured at the census block group level and then aggregated to the county and state levels to examine spatial disparities.

Accessibility Measurement

Accessibility was defined as the number of emergency mental health facilities reachable within a given travel-time threshold (Isochrone). Accessibility was first measured at the census block group level using 15-minute and 30-minute driving-time isochrones generated from real-world road networks.

$$A_i = \sum_j I(T_{ij} \leq \tau)$$

Block group-level accessibility was then:

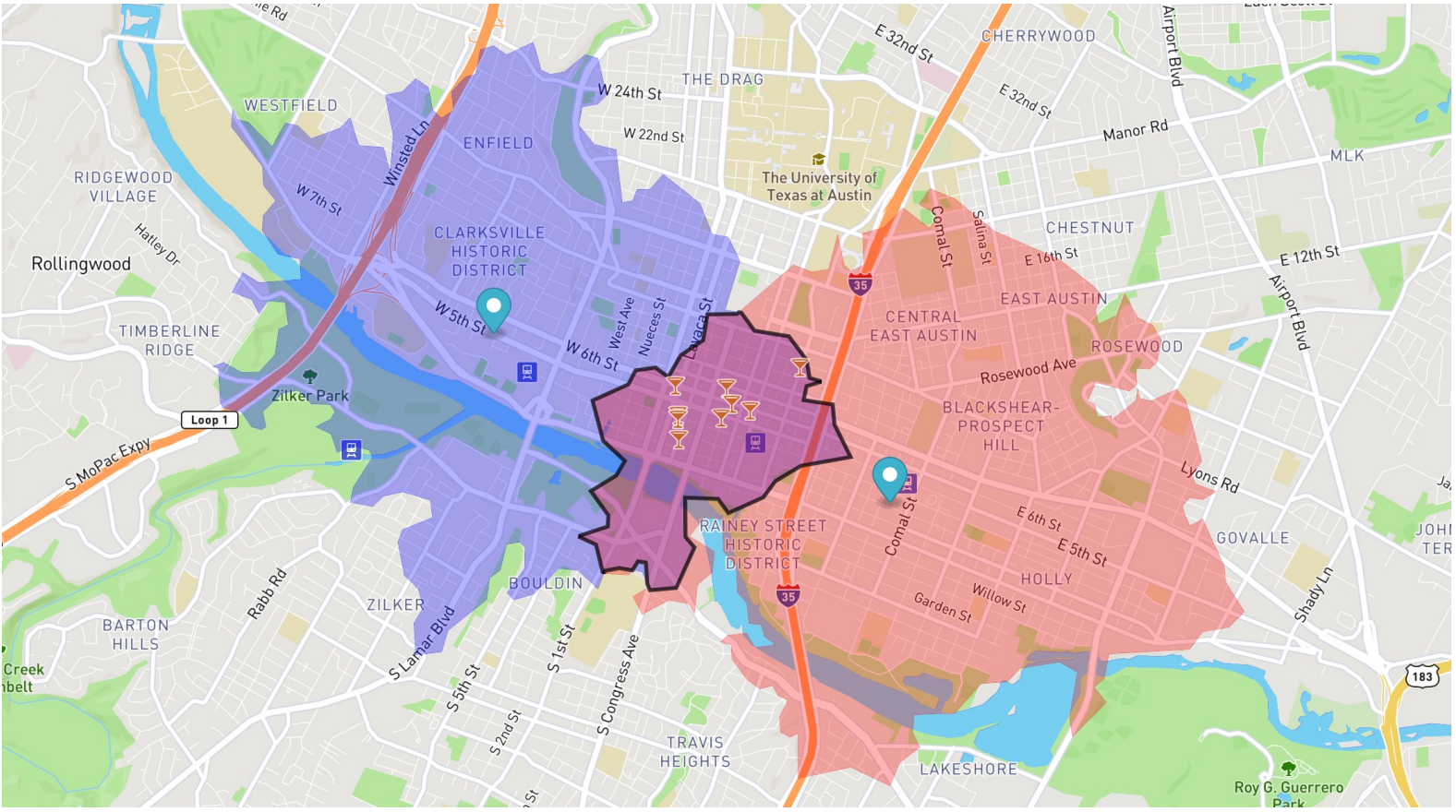
- **Aggregated to the county level** using population-weighted averages.
- **Summarized as population coverage %** representing the percentage of residents' accessibility to Emergency Mental Health services

MGWR Analysis

To examine spatially varying relationships between accessibility and socioeconomic factors, we applied Multiscale Geographically Weighted Regression (MGWR) at the county level.

Dataset

- U.S. Census Bureau TIGER/Line shapefiles (2025)
- National Mental Health Services Survey (N-MHSS)
- Mapbox Isochrone API
- American Community Survey (ACS) 2019–2023 5-year estimates:
 - Population density
 - Median household income (inflation-adjusted)
 - Educational attainment
 - Employment ratio (ages 16+)



Generating 30-min driving isochrone through Mapbox API

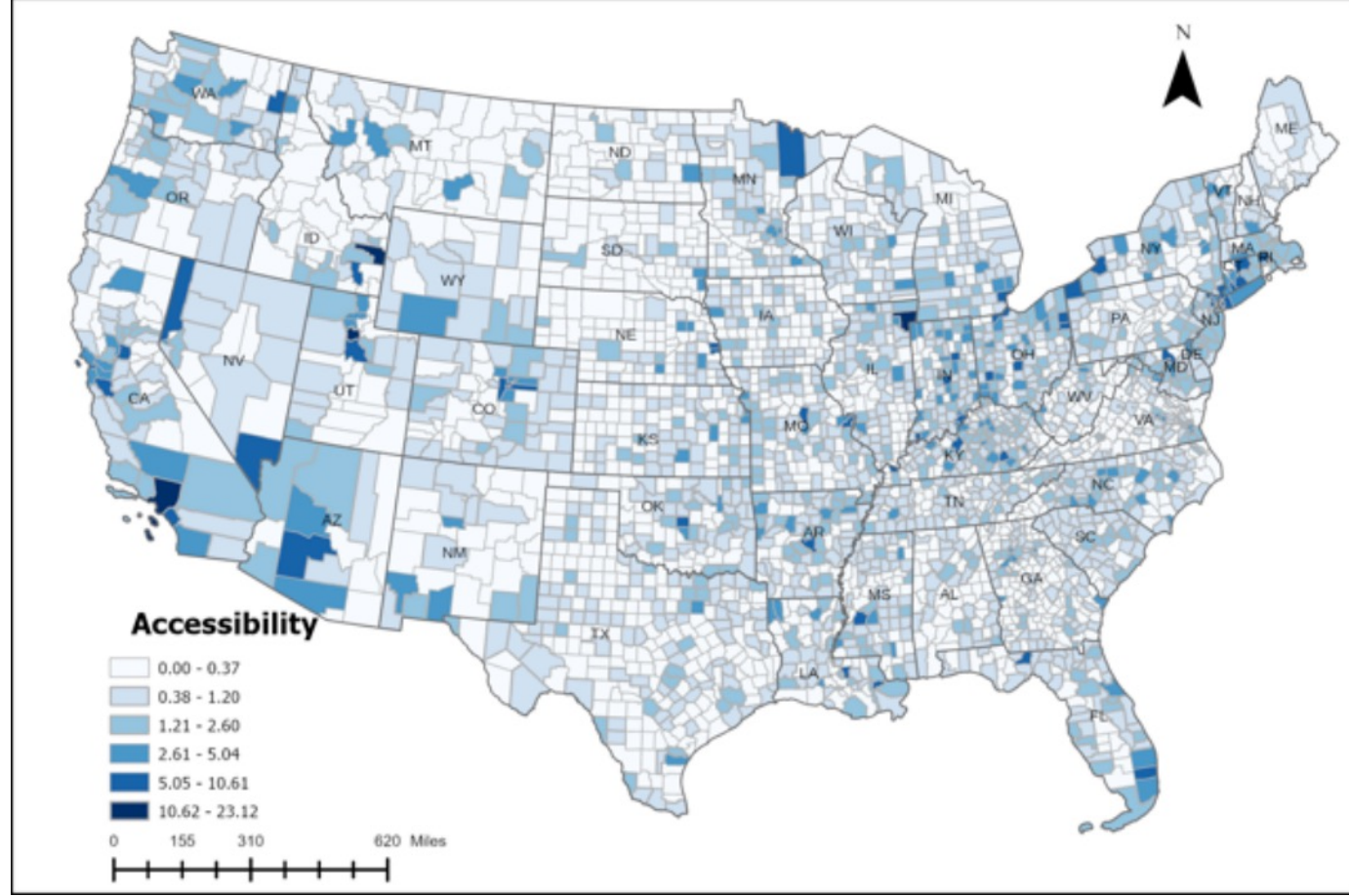
Software

- Python 3.13 – data processing and statistical analysis
- ArcGIS Pro 3.2.1 – MGWR modeling, spatial analysis, and mapping

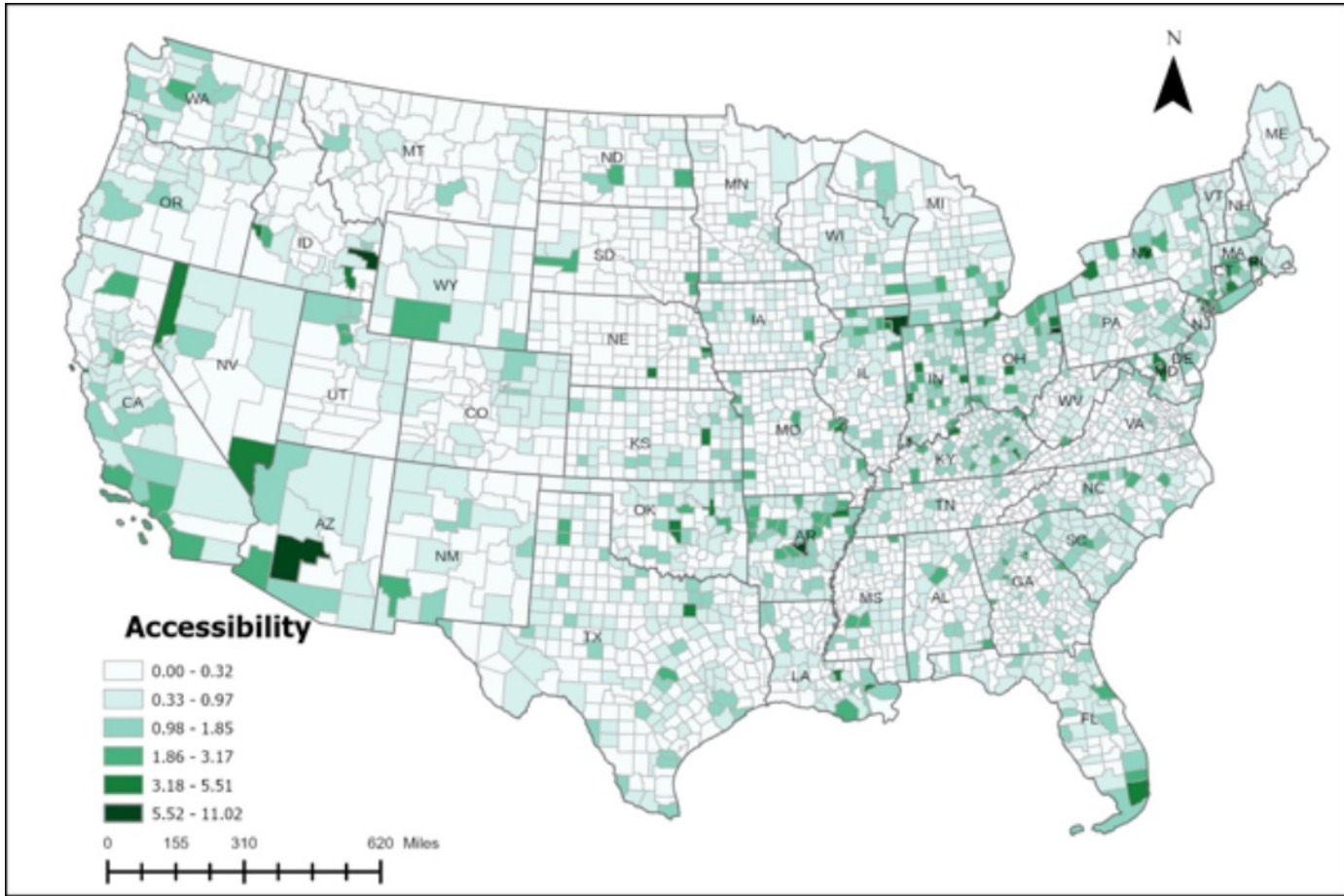
RESULTS

Spatial Accessibility Patterns

County-level maps show pronounced geographic disparities in access to emergency mental health services. Accessibility is highly concentrated in metropolitan areas, while large rural regions exhibit low or near-zero access. Crisis Intervention Teams demonstrate broader coverage than Psychiatric Emergency Walk-In Clinics, though substantial gaps persist across much of the country.



County level maps of spatial access to mental health services for Crisis Intervention Teams at 15 minutes



County level maps of spatial access to mental health services for Psychiatric Emergency Walk-In Clinics at 15 minutes

Factors Associated with Accessibility

MGWR results revealed significant spatial heterogeneity in the factors associated with emergency mental health accessibility. Population density and educational attainment were positive correlates across most counties, consistently

State-Level Variation

State	15-min All Clinics (%)	Rank	30-min All Clinics (%)	Rank
Nevada	83.8	1	91.6	24
New York	83.3	2	97.0	5
Utah	83.0	3	90.7	23
Illinois	82.1	4	97.4	8
California	81.9	5	94.1	14

Table 1. State Level Accessibility Ranking

State-level population coverage reveals strong interstate variation (Table 1). States such as Nevada, New York, Utah, Illinois, and California rank highest at the 15-minute threshold, while many rural states remain persistently underserved. Accessibility improves at 30 minutes, but disparities across service types remain.

DISCUSSIONS AND FUTURE WORK

Accessibility is highly concentrated in metropolitan and regional hubs, while many rural and peripheral counties remain persistently underserved, even under longer travel-time thresholds.

Importantly, these disparities reflect not only population distribution but also the cumulative effects of federal and state policy decisions. Federal initiatives establish baseline opportunities for crisis system expansion, yet state-level adoption, funding priorities, and regulatory frameworks play a decisive role in shaping where services are ultimately available.

Future research should incorporate service capacity measures, such as staffing levels and full-time equivalent providers, and extend accessibility modeling to include public transportation and mobile crisis services.

CONCLUSION

This study demonstrates substantial geographic variation in access to emergency mental health services across the U.S., driven largely by differences in where crisis services have been established and how state policies support their development. Access is concentrated in metropolitan areas, while many rural and peripheral communities remain underserved.

These patterns highlight the central role of policy in shaping emergency mental health systems: federal initiatives create opportunities for expansion, but state-level policy choices determine whether services are equitably distributed. Public health efforts should prioritize policy strategies that support coordinated crisis systems and direct resources to counties with persistent gaps in service availability.

ACKNOWLEDGEMENTS

The author gratefully acknowledges the guidance and mentorship of Dr. Junghwan Kim and Dr. Mengxi Zhang from Virginia Tech. This research was conducted as part of an internship with the Harvard Spatial Data Lab, whose support and research environment made this work possible.

ILLINOIS

VIRGINIA TECH

Spatial Data Lab
Internship Program